

Sagar Mandiya

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Education

BOSTON UNIVERSITY

Master of Science, Computer Science

09/2022 - 06/2024 (expected)

- Relevant Courses – Artificial Intelligence, Foundation Machine Learning, Computer Language Theory, Web Mining and Analytics.

ACROPOLIS INSTITUTE OF TECHNOLOGY AND RESEARCH

Bachelor of Technology, Computer Science and Engineering

07/2017 - 06/2021

- Relevant Courses – Object Oriented Programming & Methodology, Analysis Design of Algorithm, Pattern Recognition, Machine Learning, Theory of Computation, Internet of Things, Cloud Computing, Image Processing & Computer Vision.

Experience

BOSTON UNIVERSITY

MA, USA

Graduate Research Assistant - Bitlab@Questrom School of Business

03/23 - Present

- Conduct research on music generation using generative AI and transformers.
- Research cause and effect of generative ai in today's market with respect to content creation as a whole.

Graduate Research Assistant - BU Department of Computer Science

04/23 - Present

- Derived meaningful insights on one of the largest data available for the study on autistic children; utilised extensive and comprehensive data analysis to uncover valuable insights, focusing on autistic children and their critical age for definitive improvement saturation.
- Conducted an in-depth analysis of a robust emergency room dataset, specifically examining injuries and their causes among male and female patients. Derived significant insights into the patterns and differences between genders in terms of injury occurrences and causes in a hospital setting.

Graduate Teaching Assistant

Jan. 2023 - APR. 2023

- Conduct office hours for the class of BA820 - Unsupervised & Unstructured Machine Learning.
- Help Professor in developing learning activities to keep students actively engaged.

TENSORGO TECHNOLOGIES PRIVATE LIMITED

Hyderabad, India

Computer Vision Engineer

05/22 - 08/22

- Worked extensively on virtual perimeter overlay; computer vision project involving image segmentation and object detection.
- Developed an in-depth understanding of model tuning for optimal performance on image related problems.

Associate Computer Vision Engineer

09/21 - 05/22

- Designed project pipeline for a complex deep learning based project from scratch and implemented it effectively.
- Constructively worked with team members as a project manager and developer and increased product efficiency by 20%.
- Effectively communicated with team members and stakeholders to resolve problems at an early stage.
- Promoted within an year for excellent performance.

Publications

Neural networks with rhythmic flow* :

(2022)

- Study on enhancing neural network architecture based on human brain's capability of executing tasks on auto-pilot, like driving a car; thus achieving multi-tasking with a focus on reducing hardware resource requirements.

Deep Fake Detection: Survey of Facial Manipulation Detection Solutions: :

(2021)

Published in International Research Journal of Engineering and Technology (IRJET) (arxiv.org/abs/2106.12605)

- Studied the advances in the **Facial Manipulation** field, and trained state-of-the-art networks for **performance analysis**.

* On-Going Publication

Academic Projects

Answer Verification System - TensorGo Technologies Private Limited

-Team size: 3

- Working as a project manager on developing a **deep learning** based model for the use case of interviews, to verify the response given by interviewees and assign them a relevance score.

Automatic Speech Recognition and Voice Biometrics - TensorGo Technologies Private Limited

-Team size: 2

- Developed a **deep learning** based model to transcribe the speech and recognize the speaker based on their pre saved embeddings, using Nvidia NeMo

Face Mask and Social Distancing Detection- TensorGo Technologies Private Limited

-Team size: 1

- Designed and integrated the **operation cycle, and face detection caffe model** on stand-alone unit meant to detect if the social distancing and face mask guidelines are being followed at public places.

Deep Fake Detection - Prof. Rashid Sheikh & Prof. Vandana Kate

-Team size: 3

- Executed the **data collection, pre-processing, training, fine-tuning and analyzed** the XceptionNet, VGG-19 and Mesonet models for better performance on the use case of detecting fake images and videos circulating the internet.

Sarcasm Detection in News Headlines

-Team size: 1

- Developed a **bi-directional LSTM** based model capable of detecting sarcasm from the news headlines.

Portfolio Website and Blog

-Team size: 1

- Developed a portfolio website cum blog from scratch using HTML5, CSS3, and JAVASCRIPT, which can be visited at <https://www.sagarmandiya.com/>

Technical Skills

- **Programming Languages:** C, C++, Python, R, HTML, CSS, Javascript, PHP, SQL.
- **Libraries:** Numpy, Pandas, OpenCV, Imutils, Matplotlib, Scikit-learn.
- **Frameworks:** Nvidia CUDA, TensorRT, PyTorch, TensorFlow.
- **Operating Systems:** Ubuntu, Kali linux, MacOS, Windows.
- **Additional Skills:** WordPress, Bootstrap, Git.

Certifications

- **Deep Learning Specialization** by deeplearning.ai.
- **Machine Learning** by Stanford.
- **Convolutional Neural Networks in TensorFlow** by deeplearning.ai.
- **Introduction to TensorFlow for AI, ML, and DL** by deeplearning.ai.
- **Databases and SQL for Data Science** by IBM.
- **The Bits and Bytes of Computer Networking** by Google.
- **Python Data Structures** by University of Michigan.

* Click on course title to access the certification

Activities

- Participated in the **Red Cross Society** blood donation camp under **National Service Scheme (NSS)**, 2022.
- Organized a **charity play-zone** named "Khel-Adda" in annual university cultural festival to raise funds which were donated to the families of martyrs of 2019 Pulwama Attack, 2019.
- **Armageddon: Annual Technical Festival of AITR, Indore**, 2019.
 - Integral part of the **organizing team** of national level Tech-Fest.
 - Organized **Technical Quiz** in the Tech-Fest to encourage the students to dig deep into the field of computer science.
- Qualified for the second round of International Coding Competition organized by Codechef, 2019.